

CMLE: A Collaborative LoRA-Enhanced Expert Framework for Multimodal Fake News Detection

Quan-Yi Zou¹, Member, IEEE, Gang Chen¹, Fangyanuo Zhou, Xiao-Kun Wu², Member, IEEE, Zhan-Yu Yang³, Member, IEEE, and Yue-Yue Shi

Abstract—The rise of multimodal fake news, often combining deceptive text, manipulated images, and synthetic biometric cues such as deepfaked faces, voices, or gestures, presents serious challenges to media credibility and online trust. Existing detection models typically rely on rigid cross-modal alignment and require extensive fine-tuning, limiting their scalability and robustness. To address these challenges, we propose CMLE (Collaborative Multi-LoRA Experts), a unified and efficient framework for robust fake news detection enhanced by biometric anti-spoofing and integrity verification. CMLE deploys multiple lightweight LoRA-based expert modules, each specialized in modality-specific cues: textual semantics, visual manipulation and user feedback. The universal LoRA expert captures shared cross-modal semantics and contextual information. To align specialized and global representations, we introduce a Mutual Information Maximization (MIM) strategy that encourages semantic consistency while preserving expert diversity. Experiments on extended multimodal fake news datasets—augmented with biometric forgeries—demonstrate that CMLE achieves state-of-the-art performance in both content-level and identity-level deception detection.

Index Terms—Multimodal fake news detection, low-rank adaptation (LoRA), collaborative experts, dynamic gating mechanism, parameter-efficient fine-tuning.

I. INTRODUCTION

SOCIAL media platforms, such as Twitter, Weibo, and Tiktok, have emerged as the main avenues for the public to obtain and share information. The proliferation of multimodal fake news, which blends misleading textual narratives with manipulated images, synthetic speech, and deepfake videos, has become a significant threat to public trust and the integrity of online information ecosystems. With the rapid

advancement of fake content generation, adversaries increasingly employ biometric forgeries—such as facial manipulation, voice imitation, and behavioral spoofing—to produce convincing deceptive posts. This evolution challenges automated detection systems to assess both content authenticity and user integrity across multiple modalities. Consequently, fake news can swiftly propagate through interconnected devices, shaping public perception.

Early fake news detection (FND) relied on uni-modal methods that separately analyzed textual or visual content. These models extracted semantic, emotional, stance, and intent features from text and surface cues such as grammatical or image artifacts [1]. However, they failed to capture cross-modal inconsistencies, where genuine images may accompany deceptive text [2]. To overcome this, recent multimodal FND (MFND) studies integrate features across modalities [3], employing fusion and alignment techniques [4]—including consistency learning [5], and joint representation learning [6]—to boost detection accuracy. Despite recent advances, existing multimodal fake news detection (MFND) methods often rely on static fusion or simple concatenation, assuming strong cross-modal consistency and thus struggle to capture subtle text-image inconsistencies common in real-world scenarios. Moreover, existing methods typically rely on large-scale pretraining (such as BERT [7] or ViT [8]), and rigid cross-modal alignment, which makes them inefficient, non-adaptive, and unsuitable for on-device deployment. From a consumer technology perspective, these limitations are prominent in user-facing systems such as social media, mobile news services, and intelligent assistants in consumer electronics. Operating on resource-constrained devices, these platforms demand lightweight and adaptive detection models that are both computationally efficient and capable of capturing fine-grained cross-modal inconsistencies.

To address the limitations of existing MFND methods in terms of efficiency and adaptability, this paper proposes a novel framework named Collaborative Multi-LoRA Experts (CMLE). CMLE introduces multiple lightweight experts enhanced by Low-Rank Adaptation (LoRA), each specialized in extracting features from news comments, news text, and images. By leveraging LoRA, each expert can be efficiently fine-tuned with minimal trainable parameters, significantly reducing computational cost while maintaining strong performance. Furthermore, we design a dynamic gating mechanism that adaptively integrates predictions from different experts

Received 23 July 2025; revised 11 October 2025 and 25 January 2026; accepted 22 March 2026. Date of publication 10 April 2026; date of current version 28 August 2026. This work was supported in part by Guangdong Philosophy and Social Science Foundation Regular Project under Grant GD24YXW02 and in part by China Post-Doctoral Science Foundation under Grant 2024M750935 and Grant 2025T180948. (Corresponding author: Gang Chen.)

Quan-Yi Zou and Gang Chen are with the School of Journalism and Communication, South China University of Technology, Guangzhou 510006, China (e-mail: zouquanyi@scut.edu.cn; cgcg518@scut.edu.cn).

Fangyanuo Zhou and Yue-Yue Shi are with the School of Software, South China University of Technology, Guangzhou 510006, China (e-mail: 1966322816@qq.com; msshiyueyue@mail.scut.edu.cn).

Xiao-Kun Wu is with the Renmin University of China, Beijing 100872, China (e-mail: wuxiaokun@ruc.edu.cn).

Zhan-Yu Yang is with the School of Artificial Intelligence, Anhui University, Hefei 230601, China (e-mail: yangzhanyu@hotmail.com).

Digital Object Identifier 10.1109/TCE.2026.3677445

1558-4127 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Authorized licensed use limited to: SOUTH CHINA UNIVERSITY OF TECHNOLOGY. Downloaded on September 10, 2026 at 06:11:18 UTC from IEEE Xplore. Restrictions apply.

based on their reliability and contextual relevance, enabling robust and context-aware detection. The model learns a shared cross-modal representation, enabling seamless interaction among textual, image, and social comments for holistic understanding of misinformation. Extensive experiments on benchmark datasets demonstrate that CMLE achieves superior performance compared to state-of-the-art approaches while significantly improving efficiency. These results confirm the effectiveness of our collaborative expert design and highlight the potential of LoRA-based lightweight adaptations in multi-modal fake news detection. In summary, this work makes the following contributions:

- A novel approach, CMLE, is proposed to integrate multiple lightweight experts specialized in textual, visual, user comment and cross-modal feature extraction through LoRA.
- We propose a Mutual Information Maximization strategy to align global and local representations across modalities, improving semantic coherence and deception detection.

II. RELATED WORKS

A. Uni-Modal Fake News Detection

Early online news was predominantly text-based, with minimal visual content, leading initial fake news detection research to focus on analyzing text and images independently. Before the rise of deep learning, fake news detection (FND) relied on statistical feature engineering—such as word frequency and embeddings—combined with traditional classifiers. Ma et al. [9] introduced RNNs for automatic text representation learning, initiating the move toward end-to-end models. Subsequently, deep learning methods including CNNs [10], GCNs [11], and knowledge graphs [12] achieved notable success by capturing complex data patterns. Recent advances in fake news detection have leveraged powerful NLP models such as BART [7] and RoBERTa [13], which effectively extracted contextual dependencies among words. These transformer-based models excel at understanding subtle linguistic nuances, enabling more reliable identification of misinformation across diverse contexts.

In social networks, the assessment of the authenticity of news sometimes requires social context or relevant knowledge. Relying solely on the news content is often insufficient. Thus, integrating external knowledge into the model is necessary to enhance its detection capabilities. Dun et al. [14] incorporated knowledge graphs into news text via entity alignment and attention mechanisms to enhance fake news detection. Liao et al. [15] proposed a multi-step evidence retrieval framework simulating human verification behavior, improving both efficiency and accuracy. However, uni-modal approaches remain limited by their focus on single modalities, making them less capable of capturing cross-modal inconsistencies and more susceptible to multi-modal adversarial attacks [16].

B. Multi-Modal Fake News Detection

Multimodal approaches enhance detection by integrating complementary information from text, images, and propagation networks, leveraging subtle inter-modal associations.

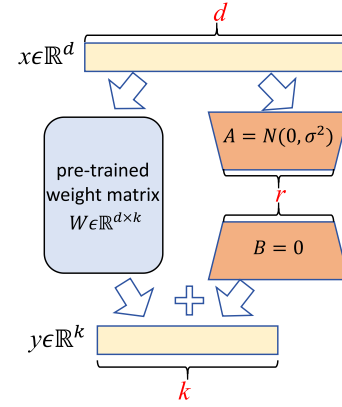


Fig. 1. LoRA in the transformer block [24].

Current MFND based on multi-modal fusion methods are categorized into two main paradigms based on feature fusion stages: early and late fusion strategies. Early fusion strategies concatenate features from different modalities before classification. To further accomplish more complex tasks, Wang et al. [17] integrated concepts of professional knowledge with original text and images by uniformly mapping information from different modalities into graph nodes prior to computational processing.

In contrast to early fusion, late fusion methods use separate feature extractors for each modality before integration. For instance, Xue et al. [18] proposed MCNN, combining BERT-based text features and ResNet-50 image features to detect cross-modal inconsistencies. Other late fusion approaches aggregate modality-specific predictions via majority voting [19]. Beyond feature fusion, some methods incorporate news propagation structures. Bian et al. [20] introduced a bidirectional GCN to capture top-down dissemination patterns. Zhang et al. [21] proposed a Bayesian edge-enhanced GCN to model relation reliability. MCAN [5] employs multi-task learning to jointly perform fake news detection and event classification, reducing event bias and enhancing generalization.

C. LoRA Model

Pretrained large-scale pretrained models underpin modern machine learning, yet full fine-tuning remains computationally costly due to their vast parameter counts. Low-Rank Adaptation (LoRA) offers an efficient alternative by freezing original weights and inserting trainable low-rank matrices into selected layers, greatly reducing parameter overhead while maintaining performance [22], [23]. As illustrated in Figure 1, the LoRA-enhanced forward computation can be expressed as:

$$y = \text{LoRA}(x) = Wx + BAx, \quad (1)$$

Formally, a weight update matrix $W \in \mathbb{R}^{d \times k}$ is reparameterized as the product of two lower-dimensional matrices, $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$, where $r \ll \min(d, k)$.

This low-rank formulation significantly reduces the number of learnable parameters from $O(dk)$ to $O(r(d+k))$, thereby improving memory efficiency and computational scalability.

Despite the compact parameterization, LoRA maintains competitive task performance, making it well-suited for large-scale and multi-task scenarios.

This formulation is applied to the query, key, and value projections as follows:

$$Q = (W_Q + B_Q A_Q)X, \quad (2)$$

$$K = (W_K + B_K A_K)X, \quad (3)$$

$$V = (W_V + B_V A_V)X, \quad (4)$$

where $X \in \mathbb{R}^{n \times d}$ is the input sequence of length n , and each pair (A_*, B_*) corresponds to a specific low-rank adapter for the respective projection. The attention output is then computed as usual:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right)V.$$

In the pre-training setting, ReLoRA [22] and MoRA [25] employ low-rank parameter updates to enhance training efficiency in high-dimensional architectures. In continual learning scenarios, several LoRA-based strategies have been introduced to mitigate catastrophic forgetting. InfLoRA [23] tackles this issue by reparameterizing pre-trained weights within a compact subspace, relying on a reduced set of parameters. GS-LoRA [26] incorporates group-wise sparse regularization to selectively preserve critical LoRA components while deactivating redundant parts, thus maintaining prior knowledge.

III. METHODOLOGY

A. Overview of CMLE

Recent advances in multimodal misinformation detection emphasize efficiency and semantic alignment, yet traditional methods often face high parameter costs and limited adaptability. To address these issues, CMLE integrates lightweight modality-specific LoRA experts with mutual-information-guided feature fusion, enabling efficient specialization for text and image cues while aligning cross-modal representations. This design enhances robustness to noisy, dynamic data and achieves competitive performance with high computational efficiency.

The proposed CMLE integrates multimodal inputs—news text, user comments, and images—into a unified architecture (Figure 2). Text and comments are tokenized and embedded to capture linguistic features, while images are processed via an adapter and CLIP vision encoder for high-level visual semantics.

The encoded features are fed into specialized LoRA experts: a Text Expert for semantic credibility, a Comment Expert for social feedback, and an Image Expert for visual manipulation cues. A universal expert captures global cross-modal representations to support all specialized branches. To align and enhance knowledge sharing, a Mutual Information Maximization (MIM) objective is applied between the universal and expert-specific representations. A Gate Router adaptively fuses outputs from all experts, and a softmax layer produces final class probabilities.

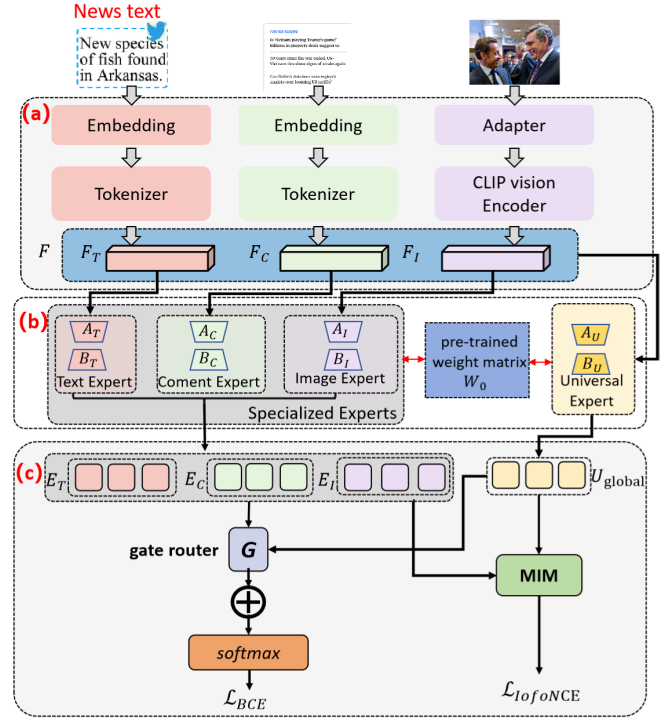


Fig. 2. The overview of the proposed CMLE framework for fake news detection. (a) Feature extraction (b) Collaboration of multiple LoRA experts (c) Feature fusion and classification.

B. Feature Encoding

This paper defines a set of multimodal news articles sample as $\mathcal{M} = \{M_1, M_2, \dots, M_n\}$, where each news article $M \in \mathcal{M}$ consists of a textual component T , an image component I , and a user comment component $C = \{c_i\}_{i=1}^N$. Here, T denotes the main news text (which can optionally include summaries or titles) and C contains related user comments that provide social feedback or reactions. The news text T and user comments C are first tokenized and then encoded using a pretrained language model (e.g. BERT) to generate contextualized semantic embeddings. Specifically, the textual representation of the news text is defined $F_T = \text{BERT}(T)$. Comment C is encoded as $F_C = \text{Aggregate}(\{\text{BERT}(c_i)\}_{i=1}^N)$. The news image I_i is passed through an adapter module to normalize low-level features, and then encoded by a vision backbone (e.g., CLIP vision encoder) to extract high-level semantic and visual manipulation signals. $F_I = \text{ImageEncoder}(\text{Adapter}(I))$ denotes the image-level or patch-level embeddings.

C. Collaborative Multi-LoRA Experts Module

Our approach aims to achieve comprehensive and robust MFND by leveraging both global and task-specific representations. This design strategically combines a universal expert with multiple specialized LoRA experts, each focusing on different aspects of the input data to enable fine-grained and complementary feature modeling.

1) *Universal Expert*: The universal expert captures shared multimodal knowledge for better alignment and generalization. Using LoRA-based low-rank adaptation, it efficiently injects

task-specific information into pretrained backbones, achieving scalable, expressive modeling with minimal parameters. The output of the universal expert U is defined as:

$$U_{\text{global}} = U\text{LoRA}(F) = W_0 F + B_u A_u F, \quad (5)$$

where $W_0 \in \mathbb{R}^{d_U \times d_F}$ denotes the pre-trained weight matrix, and $A_u \in \mathbb{R}^{r \times d_F}$, $B_u \in \mathbb{R}^{d_U \times r}$ represent the low-rank adaptation matrices specific to the universal expert. Here, r is the rank constraint with $r \ll \min(d_U, d_F)$, which ensures parameter efficiency while maintaining expressive capability. By learning only A_u and B_u , the universal expert can efficiently adapt to the task without updating the full weight matrix W_0 , thus providing a lightweight yet powerful global feature backbone that supports all specialized experts. Let U_{global} denote the global feature representation generated by the universal expert. F denotes the shared multimodal feature vector. Specifically, $F = [F_T; F_C; F_I]$ is constructed by concatenating features from different modalities (e.g., news text, comments, and images), each projected into a common embedding space.

2) *Specialized Experts*: The specialized experts extract fine-grained, modality-specific representations that complement the universal expert's global features. Through LoRA-based parameter-efficient adaptation, each expert injects task-specific knowledge into the shared backbone with minimal overhead. These experts include the text expert, image expert, and comment expert, all implemented using parameter-efficient LoRA-based adaptation. The LoRA-based transformation is defined as

$$E_m = S\text{LoRA}(F_m) = W_0 F_m + B_m A_m F_m, \quad m \in \{T, C, I\} \quad (6)$$

E_m represent the feature output from the m -th specialized expert, where $m \in \{T, C, I\}$. Each specialized expert utilizes LoRA to inject task-specific knowledge into the shared backbone with minimal trainable parameters.

D. Mutual Information Maximization (MIM)

To enhance cross-modal alignment while retaining modality-specific information, we adopt a Mutual Information Maximization (MIM) strategy that enforces semantic consistency between the global representation (U_{global}) and each expert embedding (E_m). To protect sensitive biometric data, differential privacy is integrated into the MIM process, mitigating model inversion risks and preventing reconstruction of individual features. Formally, the mutual information between U_{global} and E_m is defined as:

$$I(U_{\text{global}}, E_m) = H(U_{\text{global}}) - H(U_{\text{global}}|E_m), \quad (7)$$

where $H(\cdot)$ denotes entropy and $H(\cdot|\cdot)$ is conditional entropy. However, directly computing mutual information is intractable in high-dimensional spaces. To overcome this, we employ the InfoNCE loss as a variational lower-bound estimator, which is widely used in contrastive learning. The InfoNCE loss is formulated as:

$$\mathcal{L}_{\text{InfoNCE}} = -\mathbb{E} \left[\log \frac{e^{\text{sim}(F_{\text{global}}, E_m)/\tau}}{\sum_{E_j \in \mathcal{N}} e^{\text{sim}(F_{\text{global}}, E_j)/\tau}} \right], \quad (8)$$

where $\text{sim}(\cdot, \cdot)$ denotes a similarity function (e.g., cosine similarity), τ is a temperature parameter, and $\mathcal{N}$ is a set of negative samples drawn from other instances in the batch.

By minimizing $\mathcal{L}_{\text{InfoNCE}}$, we effectively maximize a lower bound of the mutual information between U_{global} and E_m , which helps to enforce semantic coherence while allowing each specialized expert to maintain task-specific discriminative power. To ensure semantic consistency between global and specialized representations, we use mutual information maximization. This encourages different feature representations to share more information, achieving semantic coherence across modalities while keeping expert-specific distinctions.

E. Dynamic Gating Mechanism (DGM)

To flexibly balance global and specialized feature contributions, we propose a dynamic gating mechanism that adaptively fuses outputs from the universal LoRA expert and multiple specialized LoRA experts at the token or region level. For each input token (or image patch) i , let $U_{\text{global},i}$ denote the output from the universal expert, and $E_{m,i}$ represent the output from the m -th specialized expert corresponding to modality m . The gating mechanism computes a set of adaptive weights $G_i = [g_{1,i}, g_{2,i}, \dots, g_{M+1,i}]$, which are used to dynamically control the contribution of each expert to the final representation.

Specifically, the gate weights are obtained via a linear transformation followed by a softmax operation:

$$G_i = \text{softmax}(W_g x_i + b_g), \quad (9)$$

where W_g and b_g are learnable parameters, and x_i denotes the input token feature before expert processing. The final fused representation for token i is computed as:

$$h_i = g_{1,i} \cdot F_{\text{global},i} + \sum_{m=1}^M g_{m+1,i} \cdot E_{m,i}. \quad (10)$$

The gating mechanism adaptively emphasizes the most informative experts based on input reliability, enhancing robustness and interpretability. Token-level representations h_i are aggregated into a global fused feature h_{fused} , which is passed through a softmax classifier to produce the final prediction. We aggregate them to form a global fused feature representation:

$$H_{\text{fused}} = \text{Pool}(\{h_1, h_2, \dots, h_N\}), \quad (11)$$

where $\text{Pool}(\cdot)$ denotes a pooling function, such as mean pooling. Specifically, mean pooling can be implemented as:

$$H_{\text{fused}} = \frac{1}{N} \sum_{i=1}^N h_i, \quad (12)$$

where N is the total number of tokens or patches. Finally, the fused feature is passed into an softmax classifier to generate the final prediction probability:

$$\hat{y} = \text{softmax}(H_{\text{fused}}). \quad (13)$$

Here, $\hat{y} \in [0, 1]$ denotes the probability of the input news being fake.

F. Loss Function

To effectively train the proposed CMLE model, we design a composite loss function that jointly optimizes classification performance and feature consistency across experts.

Classification Loss. We employ the standard binary cross-entropy (BCE) loss to supervise the final prediction. Given the ground-truth label $y \in \{0, 1\}$ and predicted probability $\hat{y} \in [0, 1]$, the BCE loss is defined as:

$$\mathcal{L}_{\text{BCE}} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]. \quad (14)$$

Algorithm 1 CMLE

Input: News text T , user comments $C = \{c_i\}_{i=1}^N$, news image I , label y

Output: Predicted probability $\hat{y}$

Feature Encoding:

$F_T \leftarrow \text{BERT}(T)$
 $F_C \leftarrow \text{Aggregate}(\{\text{BERT}(c_i)\}_{i=1}^N)$
 $F_I \leftarrow \text{ImageEncoder}(I)$
 $F \leftarrow [F_W; F_C; F_I]$

Universal LoRA Expert:

$U_{\text{global}} \leftarrow W_0 H + B_u A_u H$

Specialized LoRA Experts:

$E_T \leftarrow W_0 F_T + B_T A_T F_T$
 $E_C \leftarrow W_0 F_C + B_C A_C F_C$
 $E_I \leftarrow W_0 F_I + B_I A_I F_I$

Dynamic Gate:

Compute $G_i = \text{softmax}(W_g x_i + b_g)$ for each token i
 $h_i \leftarrow g_{1,i} \cdot U_{\text{global},i} + g_{2,i} \cdot E_{W,i} + g_{3,i} \cdot E_{C,i} + g_{4,i} \cdot E_{I,i}$

Global Fusion:

$H_{\text{fused}} \leftarrow \frac{1}{N} \sum_{i=1}^N h_i$

Prediction:

$\hat{y} \leftarrow \text{softmax}(H_{\text{fused}})$

Loss:

$\mathcal{L}_{\text{BCE}} \leftarrow -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$
 $\mathcal{L}_{\text{InfoNCE}} \leftarrow \text{InfoNCE loss between } U_{\text{global}} \text{ and } \{E_T, E_C, E_I\}$
 $\mathcal{L} \leftarrow \mathcal{L}_{\text{BCE}} + \lambda \cdot \mathcal{L}_{\text{InfoNCE}}$

Update:

Update all trainable parameters via back propagation

return $\hat{y}$

Overall Loss. The total loss for training the CMLE framework combines these components:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \cdot \mathcal{L}_{\text{InfoNCE}}, \quad (15)$$

where λ is a balancing hyperparameter that controls the weight of the consistency regularization term. Algorithm 1 outlines the detailed procedure.

IV. EXPERIMENTS

A. Experimental Settings

We evaluate CMLE on three benchmark multimodal fake news detection datasets—Weibo, Twitter, and DGM⁴ [27]—each containing verified text-image pairs. Text is encoded using BERT-base, and images are processed via CLIP ViT-B/32. Both modalities output 512-dimensional features, with the LoRA rank set to 8 for parameter efficiency.

All models are trained for 10 epochs using the Adam optimizer with a learning rate of 1×10^{-4} and batch size 32. The mutual information regularization weight λ is set to 1. Training is conducted on an NVIDIA RTX 4060 GPU. We evaluate performance using Accuracy, Precision, Recall, and F1-score, following standard protocols.

B. Comparison Baselines

To evaluate CMLE comprehensively, we compare it with representative multimodal fake news detection baselines. SpotFake [10] combines text encoders and CNN-based visual models, while SpotFake+ [28] enhances contextual understanding with XLNet. SAFE [29] introduces similarity-aware multimodal consistency modeling, and MCAN [5] applies co-attention to jointly process text and images. EANN [30] employs multi-task learning with event classification to reduce bias, and FND-CLIP [31] aligns modalities via CLIP embeddings and attention-based fusion (with identical preprocessing for Chinese datasets). MIMoE-FND [32] uses a hierarchical Mixture-of-Experts and gating mechanism for adaptive fusion, while MFIR [33] detects misinformation through cross-modal inconsistency reasoning. These comparisons highlight CMLE's superior ability to capture cross-modal dependencies and achieve robust detection performance.

C. Analysis of Results

Table I summarizes the performance of CMLE and baseline methods on the Weibo, Twitter, and DGM⁴ datasets. CMLE consistently achieves the highest accuracy—92.9%, 93.4%, and 93.5%—outperforming the best baseline by 2.2%, 0.9%, and 3.0%, respectively. It also attains the best F1-scores across all datasets, with a notable 5.3% gain over the strongest baseline on Twitter. These results confirm CMLE's superiority in both fake and real news classification tasks.

On Weibo, CMLE achieves the highest fake news recall and F1-score, with strong real news performance, showing excellent generalization. On Twitter, MFIR has competitive fake news precision, but CMLE leads in overall F1 and recall. Across all benchmarks, CMLE consistently attains the best precision and recall for true news, confirming its robustness and balanced detection capability. Compared to traditional multimodal methods (SpotFake, SAFE, MCAN), the integration of collaborative LoRA experts and dynamic gating significantly improves semantic alignment and representation quality, while mutual information maximization further enhances cross-modal consistency and generalization.

D. Ablation Analysis

To systematically evaluate the contributions of each key module in our CMLE approach, we perform comprehensive ablation experiments. We consider seven variant ablation models:

(1) **w/o All Experts** refers to the backbone encoder operating without any LoRA expert modules.

(2) **Only Text Expert**, which retains only the text-specific expert to assess the role of textual semantics in fake news detection;

TABLE I
PERFORMANCE COMPARISON WITH BASELINES ON WEIBO, TWITTER, AND DGM⁴ DATASETS

Dataset	Methods	Accuracy	Fake News			True News		
			Precision	Recall	F1-score	Precision	Recall	F1-score
Weibo	spotFake	0.892	0.902	0.932	0.917	0.847	0.656	0.786
	spotFake+	0.870	0.887	0.849	0.868	0.855	0.932	0.873
	SAFE	0.763	0.833	0.659	0.736	0.717	0.868	0.785
	MCNN	0.878	0.904	0.862	0.882	0.870	0.905	0.887
	EANN	0.827	0.847	0.812	0.829	0.807	0.843	0.825
	MVAE	0.745	0.801	0.719	0.758	0.689	0.777	0.730
	FND-CLIP	0.907	0.924	0.901	0.908	0.914	0.901	0.907
	MIMoE-FND	0.904	0.942	0.913	0.928	0.913	0.942	0.927
	MFIR	0.858	0.850	0.848	0.849	0.867	0.871	0.869
	CMLE	0.929	0.915	0.969	0.936	0.915	0.954	0.956
Twitter	spotFake	0.777	0.751	0.900	0.812	0.832	0.606	0.701
	spotFake+	0.790	0.793	0.827	0.821	0.782	0.747	0.766
	SAFE	0.766	0.777	0.795	0.786	0.752	0.731	0.742
	MCNN	0.845	0.876	0.833	0.854	0.810	0.767	0.782
	EANN	0.795	0.806	0.795	0.800	0.752	0.793	0.804
	MVAE	0.824	0.854	0.795	0.823	0.800	0.875	0.837
	FND-CLIP	0.885	0.762	0.901	0.826	0.899	0.912	0.905
	MIMoE-FND	0.895	0.906	0.644	0.698	0.953	0.936	0.936
	MFIR	0.925	0.901	0.839	0.869	0.902	0.870	0.876
	CMLE	0.934	0.978	0.864	0.875	0.926	0.952	0.952
DGM ⁴	spotFake	0.803	0.745	0.667	0.704	0.769	0.669	0.718
	spotFake+	0.800	0.730	0.668	0.697	0.832	0.869	0.850
	SAFE	0.811	0.827	0.689	0.751	0.877	0.940	0.875
	MCNN	0.858	0.812	0.734	0.770	0.829	0.876	0.852
	EANN	0.838	0.812	0.714	0.759	0.834	0.825	0.815
	MVAE	0.652	0.540	0.717	0.558	0.652	0.746	0.695
	FND-CLIP	0.825	0.651	0.770	0.704	0.799	0.836	0.802
	MIMoE-FND	0.825	0.770	0.744	0.698	0.838	0.820	0.826
	MFIR	0.905	0.901	0.819	0.839	0.846	0.803	0.806
	CMLE	0.935	0.911	0.849	0.869	0.906	0.848	0.876

(3) **Only Comment Expert**, which preserves only the comment expert to evaluate the effect of social feedback signals;

(4) **Only Image Expert**, which uses only the image expert to analyze the contribution of visual cues;

(5) **w/o Universal Expert**, which removes the universal LoRA expert to examine its role in providing global cross-modal representations;

(6) **w/o Specialized Experts**, which eliminates all task-specific experts to assess the impact of fine-grained feature modeling;

(7) **w/o MIM**, which omits the mutual information objective to validate its contribution to semantic consistency;

(8) **w/o DGM**, which replaces the dynamic gate with a fixed equal-weight fusion to analyze its adaptive fusion capability.

To evaluate the effectiveness of each component within our proposed CMLE framework, we conduct extensive ablation studies across three benchmark datasets: Weibo, Twitter, and DGM⁴. The results are presented in Table II, which reports accuracy, precision, recall, and F1-scores for both fake news and true news across various model variants.

Notably, the complete CMLE model consistently achieves the best overall performance on all metrics and datasets. For example, on the Twitter dataset, CMLE reaches an F1-score of 0.875 for fake news and 0.952 for true news. In contrast, removing all experts leads to significant degradation (Fake-F1 drops to 0.667), highlighting the crucial role of the expert modules. Furthermore, the Universal Expert and Specialized Experts contribute substantially to cross-modal learning, as

evidenced by performance drops when either is removed. MIM and the DGM also prove vital: removing them results in F1-score declines of over 3%–5% on average.

Table III further quantifies the impact by showing the maximum degradation percentages across all datasets. Removing all experts results in the most severe drop, with Fake-F1 decreasing by up to 33.01% and Real-F1 by 50.20%. Even single-modality experts (e.g., only text or image) yield large degradation, confirming the importance of multi-expert collaboration and multimodal fusion.

These results validate that each component—LoRA-based experts, universal expert, mutual information alignment, and gating mechanism—plays a complementary and indispensable role in the overall framework.

E. Parameter Analysis

To evaluate the effect of the hyperparameter λ , which controls the weight of the auxiliary mutual information maximization (MIM) term in our overall loss function, we conduct sensitivity analysis on three datasets: Weibo, Twitter, and DGM⁴. The evaluation considers three key metrics: Accuracy, F1 score for fake news (Fake-F1), and F1 score for real news (Real-F1).

As illustrated in Figure 3, each row corresponds to a dataset and each column to a metric. The dashed horizontal lines denote the baseline performance when $\lambda = 0$, i.e., with the MIM component removed. Across all datasets,

Accuracy improves as λ increases from 0.1 to approximately 1.2, after which the gain saturates or slightly declines. This

TABLE II
ABLATION STUDY RESULTS OF DIFFERENT CMLE VARIANTS ACROSS THREE DATASET

Dataset	Variants	Accuracy	Fake News			True News		
			Precision	Recall	F1-score	Precision	Recall	F1-score
Weibo	w/o All Experts	0.629	0.612	0.435	0.627	0.572	0.568	0.476
	Only Text Expert	0.630	0.619	0.410	0.768	0.525	0.696	0.516
	Only Comment Expert	0.785	0.803	0.668	0.719	0.671	0.846	0.636
	Only Image Expert	0.756	0.745	0.778	0.817	0.810	0.516	0.629
	w/o Universal Expert	0.871	0.721	0.776	0.732	0.850	0.726	0.777
	w/o Specialized Experts	0.784	0.659	0.610	0.686	0.557	0.776	0.788
	w/o MIM	0.873	0.861	0.859	0.769	0.769	0.891	0.874
	w/o DGM	0.837	0.834	0.812	0.835	0.835	0.843	0.825
	CMLE(full)	0.929	0.915	0.969	0.936	0.915	0.954	0.956
Twitter	w/o All Experts	0.642	0.625	0.745	0.627	0.687	0.678	0.676
	Only Text Expert	0.649	0.693	0.807	0.731	0.786	0.786	0.776
	Only Comment Expert	0.788	0.754	0.789	0.763	0.750	0.788	0.651
	Only Image Expert	0.725	0.796	0.749	0.734	0.848	0.647	0.656
	w/o Universal Expert	0.787	0.872	0.732	0.680	0.751	0.655	0.765
	w/o Specialized Experts	0.728	0.794	0.704	0.669	0.827	0.812	0.797
	w/o MIM	0.750	0.762	0.418	0.721	0.808	0.889	0.821
	w/o DGM	0.815	0.872	0.764	0.798	0.831	0.821	0.916
	CMLE(full)	0.934	0.978	0.864	0.875	0.926	0.915	0.952
DGM ⁴	w/o All Experts	0.752	0.665	0.744	0.674	0.767	0.678	0.735
	Only Text Expert	0.735	0.716	0.771	0.648	0.803	0.826	0.634
	Only Comment Expert	0.728	0.642	0.635	0.629	0.867	0.813	0.735
	Only Image Expert	0.734	0.705	0.539	0.650	0.682	0.824	0.806
	w/o Universal Expert	0.866	0.722	0.842	0.821	0.741	0.782	0.815
	w/o Specialized Experts	0.750	0.769	0.723	0.701	0.660	0.844	0.770
	w/o MIM	0.798	0.507	0.540	0.663	0.581	0.654	0.627
	w/o DGM	0.867	0.688	0.611	0.713	0.669	0.688	0.861
	CMLE(full)	0.935	0.911	0.849	0.869	0.906	0.848	0.876

TABLE III
MAXIMUM DEGRADATION RANGES (%) OF ABLATION VARIANTS COMPARED TO CMLE ACROSS THREE DATASETS

Variants	Acc ↓	Fake-F1 ↓	Real-F1 ↓
w/o All Experts	19.57% – 32.29%	22.44% – 33.01%	16.10% – 50.20%
Only Text Expert	21.37% – 32.19%	16.46% – 25.43%	18.49% – 46.03%
Only Comment Expert	15.48% – 22.12%	12.80% – 27.59%	16.07% – 33.46%
Only Image Expert	18.61% – 22.41%	12.72% – 25.15%	7.99% – 31.07%
w/o Universal Expert	7.39% – 15.74%	5.54% – 22.29%	6.96% – 19.67%
w/o Specialized Experts	15.67% – 22.02%	19.28% – 26.71%	12.16% – 17.64%
w/o MIM	6.03% – 19.73%	17.86% – 23.76%	8.57% – 28.39%
w/o DGM	7.27% – 12.73%	8.81% – 17.95%	1.71% – 13.70%

indicates that moderate regularization from the MIM objective promotes better generalization, whereas overly strong alignment may suppress useful modality-specific information. The Fake-F1 score rises significantly with increasing λ , reaching its peak near $\lambda \approx 1.2$ on the Weibo dataset, suggesting that semantic alignment enhances the model's sensitivity to deceptive cues. However, excessively large λ values can lead to performance degradation, likely due to overfitting to aligned features. In contrast, Real-F1 demonstrates a more stable upward trend as λ increases, especially on Twitter and DGM⁴, with the best performance typically observed in the range of [1.0, 1.6]. This implies that well-aligned multimodal representations also help the model identify truthful content more accurately. Across all datasets and metrics, the baseline performance at $\lambda = 0$ is consistently inferior, confirming the critical contribution of the auxiliary MIM loss. In conclusion, setting λ within the range of 1.0 to 1.4 offers an effective balance between semantic alignment and task-specific discrimination, leading to enhanced robustness and improved generalization of the CMLE.

F. Case Study

To further demonstrate the effectiveness of our CMLE model in identifying nuanced multimodal inconsistencies, we present a representative case study in Figure 4. Although the image appears visually plausible, cues embedded in facial image features reveal potential inconsistencies. The facial attributes of two political figures in the image are semantically unchanged at the pixel level, yet the manipulated context alters their perceived emotional states. The individual on the left, originally smiling, appears more serious, while the one on the right shifts from cheerful to contemplative. These perceptual changes are induced by red bounding boxes and contextually misleading captions—techniques that exploit biometric perception to reinforce a fabricated narrative. This case illustrates a semantic framing attack—a type of biometric spoofing that manipulates identity, expression, or intent through subtle multimodal perturbations rather than synthetic generation. The original post described a diplomatic meeting, while the altered version distorted key phrases (“are to meet” → “was forced to resign,” “Brussels” → “Paris,” and

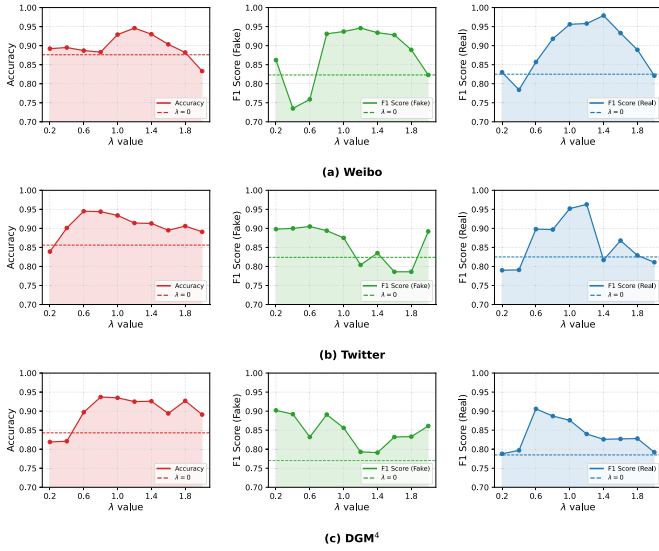


Fig. 3. Performance metrics (accuracy, fake news F1 score, real news F1 score) with different λ values.



Fig. 4. Case study of CMLE.

“friction” → “confrontation”), fabricating a political crisis. Our CMLE model, equipped with collaborative LoRA experts and mutual information regularization, accurately detects such inconsistencies by integrating facial semantics, textual tone, and cross-modal attention, outperforming unimodal baselines in identifying emotionally manipulative misinformation.

V. CONCLUSION

In this work, we propose CMLE, a novel framework that integrates collaborative LoRA experts with mutual information regularization for robust multimodal fake news detection and biometric anti-spoofing. By dynamically weighting global and domain-specific experts, CMLE effectively models subtle semantic discrepancies between textual narratives and visual cues. Furthermore, incorporation of biometric consistency—particularly facial expression semantics—enables the model to identify framing-based manipulations that exploit emotional and perceptual biases.

Extensive experiments across multiple datasets demonstrate the superiority of CMLE over strong baselines in both detection accuracy and parameter efficiency. Case study analyses further highlight CMLE’s capacity to uncover visually plausible yet semantically deceptive content, underscoring its value

in real-world misinformation scenarios where adversaries often rely on subtle biometric spoofing tactics rather than overt image tampering.

Despite its effectiveness, CMLE has certain limitations. First, the current framework focuses primarily on static multimodal inputs (text, image, comments), and does not yet address temporally evolving misinformation, such as fake news videos or time-sensitive narratives. Second, CMLE does not explicitly model fine-grained behavioral or micro-expression cues, which may carry subtle but critical deception signals. Additionally, while CMLE achieves strong performance in benchmark evaluations, its robustness under real-world deployment scenarios with noisy or missing modalities remains to be fully validated.

Future work will explore deploying CMLE in practical, user-facing consumer electronics applications to enhance the identification and prevention of misinformation, including social media content moderation systems and consumer-grade devices operating on mobile or edge platforms.

REFERENCES

- [1] T. M. H. Gedara, V. Loia, and S. Tomasiello, “Using fuzzy transform for sustainable fake news detection,” *Appl. Soft Comput.*, vol. 151, Jan. 2024, Art. no. 111173. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1568494623011912>
- [2] H. T. Phan, N. T. Nguyen, and D. Hwang, “Fake news detection: A survey of graph neural network methods,” *Appl. Soft Comput.*, vol. 139, May 2023, Art. no. 110235.
- [3] J. Hua, X. Cui, X. Li, K. Tang, and P. Zhu, “Multimodal fake news detection through data augmentation-based contrastive learning,” *Appl. Soft Comput.*, vol. 136, Mar. 2023, Art. no. 110125.
- [4] Y. Yang, J. Liu, Y. Yang, and L. Cen, “Dual-stream fusion network with multi-head self-attention for multi-modal fake news detection,” *Appl. Soft Comput.*, vol. 167, Dec. 2024, Art. no. 112358.
- [5] Y. Wu, P. Zhan, Y. Zhang, L. Wang, and Z. Xu, “Multimodal fusion with co-attention networks for fake news detection,” in *Proc. Findings Assoc. Comput. Linguistics, ACL-IJCNLP*, 2021, pp. 2560–2569.
- [6] F. Wu, S. Chen, G. Gao, Y. Ji, and X. Jing, “Balanced multi-modal learning with hierarchical fusion for fake news detection,” *Pattern Recognit.*, vol. 164, Sep. 2025, Art. no. 111485.
- [7] S. A. Aljawarneh and S. A. Swedat, “Fake news detection using enhanced BERT,” *IEEE Trans. Computat. Social Syst.*, vol. 11, no. 4, pp. 4843–4850, Aug. 2024.
- [8] E. Vishnuthirth, B. Premjith, and V. Sowmya, “Multimodal fake news prediction using a two-transformer architecture approach with llama,” in *Proc. IEEE Int. Conf. Interdiscipl. Approaches Technol. Manage. Social Innov. (IATMSI)*, Mar. 2025, pp. 1–6.
- [9] J. Ma et al., “Detecting rumors from microblogs with recurrent neural networks,” in *Proc. 25th Int. Joint Conf. Artif. Intell.*, 2016, pp. 3818–3824.
- [10] S. Singhal, R. R. Shah, T. Chakraborty, P. Kumaraguru, and S. Satoh, “SpotFake: A multi-modal framework for fake news detection,” in *Proc. IEEE 5th Int. Conf. Multimedia Big Data (BigMM)*, Sep. 2019, pp. 39–47.
- [11] I. Varlamis, D. Michail, F. Glykou, and P. Tsantilas, “A survey on the use of graph convolutional networks for combating fake news,” *Future Internet*, vol. 14, no. 3, p. 70, Feb. 2022.
- [12] B. Xie, X. Ma, J. Wu, J. Yang, and H. Fan, “Knowledge graph enhanced heterogeneous graph neural network for fake news detection,” *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2826–2837, Feb. 2024.
- [13] A. J. Keya, M. A. H. Wadud, M. F. Mridha, M. Alatiyyah, and M. A. Hamid, “AugFake-BERT: Handling imbalance through augmentation of fake news using BERT to enhance the performance of fake news classification,” *Appl. Sci.*, vol. 12, no. 17, p. 8398, Aug. 2022.
- [14] Y. Dun, K. Tu, C. Chen, C. Hou, and X. Yuan, “KAN: Knowledge-aware attention network for fake news detection,” in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 1, pp. 81–89.
- [15] H. Liao et al., “MUSER: A multi-step evidence retrieval enhancement framework for fake news detection,” in *Proc. 29th ACM SIGKDD Conf. Knowl. Discovery Data Mining*, Aug. 2023, pp. 4461–4472.

- [16] T. Zhang, "TMRB: Trusted multimedia scheme with redactable blockchain," *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 3099–3107, May 2025.
- [17] Y. Wang, S. Qian, J. Hu, Q. Fang, and C. Xu, "Fake news detection via knowledge-driven multimodal graph convolutional networks," in *Proc. Int. Conf. Multimedia Retr.* New York, NY, USA: Association for Computing Machinery, Jun. 2020, pp. 540–547.
- [18] J. Xue, Y. Wang, Y. Tian, Y. Li, L. Shi, and L. Wei, "Detecting fake news by exploring the consistency of multimodal data," *Inf. Process. Manage.*, vol. 58, no. 5, Sep. 2021, Art. no. 102610.
- [19] P. Meel and D. K. Vishwakarma, "HAN, image captioning, and forensics ensemble multimodal fake news detection," *Inf. Sci.*, vol. 567, pp. 23–41, Aug. 2021.
- [20] T. Bian et al., "Rumor detection on social media with bi-directional graph convolutional networks," in *Proc. AAAI*, vol. 34, 2020, pp. 549–556.
- [21] G. Zhang, S. Zhang, and G. Yuan, "Bayesian graph local extrema convolution with long-tail strategy for misinformation detection," *ACM Trans. Knowl. Discovery Data*, vol. 18, no. 4, pp. 1–21, May 2024.
- [22] V. Lialin, N. Shivagunde, S. Muckatira, and A. Rumshisky, "ReLoRA: High-rank training through low-rank updates," 2023, *arXiv:2307.05695*.
- [23] Y.-S. Liang and W.-J. Li, "InfLoRA: Interference-free low-rank adaptation for continual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 23638–23647.
- [24] J. E. Hu et al., "LoRA: Low-rank adaptation of large language models," 2021, doi: [10.48550/arXiv.2106.09685](https://doi.org/10.48550/arXiv.2106.09685).
- [25] T. Jiang et al., "MoRA: High-rank updating for parameter-efficient fine-tuning," 2024, *arXiv:2405.12130*.
- [26] H. Zhao et al., "Continual forgetting for pre-trained vision models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 28631–28642.
- [27] R. Shao, T. Wu, J. Wu, L. Nie, and Z. Liu, "Detecting and grounding multi-modal media manipulation and beyond," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5556–5574, Aug. 2024.
- [28] S. Singhal, A. Kabra, M. Sharma, R. R. Shah, T. Chakraborty, and P. Kumaraguru, "SpotFake: A multimodal framework for fake news detection via transfer learning (student abstract)," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 10, pp. 13915–13916.
- [29] X. Zhou, J. Wu, and R. Zafarani, "Similarity-aware multi-modal fake news detection," in *Proc. Pacific Asia Conf. Knowl. Discovery Data Mining*. Cham, Switzerland: Springer, 2020, pp. 354–367.
- [30] Y. Wang et al., "EANN: Event adversarial neural networks for multi-modal fake news detection," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, New York, NY, USA, Jul. 2018, pp. 849–857.
- [31] Y. Zhou, Y. Yang, Q. Ying, Z. Qian, and X. Zhang, "Multimodal fake news detection via CLIP-guided learning," in *Proc. IEEE Int. Conf. Multimedia Expo. (ICME)*, Jul. 2023, pp. 2825–2830.
- [32] Y. Liu, Y. Liu, Z. Li, R. Yao, Y. Zhang, and D. Wang, "Modality interactive mixture-of-experts for fake news detection," in *Proc. ACM Web Conf.*, Apr. 2025, pp. 5139–5150.
- [33] L. Wu, Y. Long, C. Gao, Z. Wang, and Y. Zhang, "MFIR: Multimodal fusion and inconsistency reasoning for explainable fake news detection," *Inf. Fusion*, vol. 100, Dec. 2023, Art. no. 101944.



the National Social Science Fund.

Gang Chen received the Ph.D. degree from Wuhan University, Wuhan, China, in 2007. He is currently a Professor with the School of Journalism and Communication, South China University of Technology, and the Executive Dean of the International Institute for Communication in Guangdong–Hong Kong–Macao Greater Bay Area. His main research areas include media sociology, digital media studies, and intelligent communication. He has led more than 20 projects at the provincial and ministerial level or above, including major, key, and general projects of



Fangyanuo Zhou is currently pursuing the bachelor's degree with the School of Software, South China University of Technology, with research methods in fake news detection and social computing.



models and mass communications.

Xiao-Kun Wu (Member, IEEE) received the Ph.D. degree in communication from Shanghai University, Shanghai, China, in 2010, and the Ph.D. degree from the School of Computer Science and Engineering, South China University of Technology, Guangzhou, China, in 2022. She is currently a Professor with the School of Journalism and Communication, Renmin University of China, Beijing, China. She has authored three academic books and over 40 research papers in her research areas. Her research interests include online collective actions and communication



Quan-Yi Zou (Member, IEEE) received the Ph.D. degree from the School of Computer Science and Engineering, South China University of Technology, Guangzhou, China, in 2022. Currently, he is an Associate Professor with the School of Journalism and Communication, South China University of Technology. He has authored over ten research papers in his research areas. His current research interests include social media, LLMs value alignment, fake news detection, and software defect prediction.

Zhan-Yu Yang (Member, IEEE), photograph and biography not available at the time of publication.

Yue-Yue Shi, photograph and biography not available at the time of publication.